INCH-POUND

MIL-F-48877A (AR)
13 October 1994
SUPERSEDING
MIL-F-48877 (AR)
17 September 1982

MILITARY SPECIFICATION

FUZE, ROCKET: M439 PARTS AND ASSEMBLY WITH DETONATOR

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, examinations and tests for the fabrication. of parts and the manufacture, assembly and packing of Fuze Rocket: M439, for the 2.75 Inch Rocket System (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise Specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-D-45495 - Detonator, Electric M84, Parts for and Loading, Assembling and Packing

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	-	Quality Assurance Terms and Definitions
MIL-STD-331	-	Fuze and Fuze Components, Environmental and Performance Tests for
MIL-STD-1167-		Ammunition Data Card

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from DODSSP - Customer Service, Standardization Documents Order Desk, 700 Robbins Avenue, Bldg . 4D, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (see 6.5)

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

PRODUCT AND PACKAGE DRAWINGS

8799112	-	Marking Diagram and Sealing for Interplant or CONUS Shipment of Ammunition Fiberboard Boxes
9260704		Fuze, Rocket, M439
9362895		Warhead 2.75 Inch Rocket, Test Empty, Modified, PXR6358 w/Flash for Test Fuze, Rocket: M439
9357775		Packing and Marking of Inner Fiberboard Box for Fuze, Rocket: M439,
9357776		Packing and Marking of Outer Fiberboard Box for Fuze, Rocket M439
9324106		Rocket Management System, Inventory - Deployment, M138
233AS431		Rocket Motor, Inert, MK66

CODE OF FEDERAL REGULATIONS

Title 49	Transportation, parts 100-199

(Copies of other Government documents, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained. (See contract provisions for additional precedence criteria.)

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with applicable drawings and specifications.

3.2 First article inspection. This specification provides for a first article inspection. Requirements for the submission of the first article sample by the contractor shall be as specified in the contract.

3.2.1 Process specification plastic parts. When a multiple cavity mold is used in the fabrication of plastic components, five (5) components from each mold cavity shall be included in the first article sample. The components shall bear a mark which identifies the component with the individual cavities.

3.3 Molded or plastic components. All molded and plastic components shall be in accordance with applicable drawings and specifications.

3.4 Fuze assemblies and parts. The fuze assemblies and parts shall comply with all requirements specified on Drawing 9260004 and associated drawings and with all requirements specified in applicable standards and specifications.

3.5 Spring embrittlement . None of the springs used in the fabrication of the Safety and Arming Device (S&A) shall break when extended to the point of minimum permanent distortion.

3.6 Spring set. The setback spring, after being subjected to a temperature of 175°F - 180°F at a compressed length of .750 inch for 24 hours and then allowed to cool at ambient for 24 hours, shall return to within .015 inch of its original free length.

3.7 Setback weight return capability. The setback weight in the S&A, after being bottomed to the base of the rotor housing by a force of 20 pounds applied equally to its two top surfaces and then released, shall return in a continuous motion to the fully safe position of locking the rotor.

3.8 Rotor reset capability The rotor assembly in the S&A, after being rotated between 26 degrees and 35 degrees from the safe position and then released, shall without external assistance and in a continuous motion be returned by the spring loaded setback weight to its fully safe and locked position.

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3.9 Separation of annular gear from number 2 sear & pinion assembly. The separation distance between the annular gear and number 2 gear and pinion, when the rotor and number 2 gear are at their extreme separated position in the S&A, shall be .038 inch maximum.

3.10 Alignment of detonator cavity. The detonator cavity. when the rotor is in the free armed detented position, shall be in line with the outside diameter of the rotor housing within .060 in. of Total Indicator Reading (TIR) functionally.

3.11 Continuity and insulation resistance. The S&A assembly, shall satisfy the continuity and insulation requirements specified in Table I of this specification.

TABLE I. S&A continuity and insulation requirements.

Position of S&A Assembly	Terminals	Requirements
Safe	#1 & #5 (Charge Switch)	30 Megohms Min. 500 VDC
Safe	#1 & #5 (Charge Switch)	100 Milliohms Max.
Armed	#1 & #5 (Charge Switch)	30 Megohms Min. 500 VDC
Armed	#1 & #2 (Charge Switch)	100 Milliohms Max.
Armed	#4 & #6 and S&A Housing	100 Milliohms Max.

3.12 Detonators. Detonators, prior to assembling to rotors, shall meet the requirements of MIL-D-45495.

3.13 Loaded S&A assembly. Appropriate safety precautions shall be taken in testing for the following requirements.

3.13.1 Load test of detonator. The detonator shall not move more than. 005 inch when a push out force of 5 pounds is applied to the detonator socket.

3.13.2 Continuity through detonator (armed). The resistance measured between terminals 3 and 4 shall be 3.5 ± 1.5 ohms.

3.13.3 Non arming. The rotor assembly in the loaded S&A shall not arm when subjected to an arming acceleration of 12.0 ± 0.5 g's minimum for a period of 5 seconds. This test may be performed with a simulated detonator.

3.13.4 S&A arming. The S&A assembly, after being temperature conditioned, shall meet the following requirements.

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3 .13.4.1 Ambient temperature arming. The S&A assembly, after conditioning at a temperature of 70 ± 15°F for 2 hours minimum, shall arm between 0.70 and 0.90 second when subjected to an arming acceleration of 55.0 ± 0.5 g.

3 .13.4.2 Low temperature arming. The S&A assembly after being conditioned at a temperature of -50 ± 5°F for 3 hours minimum, shall arm between 0.70 and 0.90 second when subjected to an arming acceleration of 55.0 ± 0.5 g.

3 .13.4.3 High temperature arming. The S&A assembly after being conditioned at a temperature of +150°F ± 5°F for 3 hours minimum shall arm between 0.70 and 0.90 second when subjected to an arming acceleration of 55.0 ± 0.5 g.

3.14 Molded Electronics assembly (Module time delay function). The module shall function within ± 2 .0 percent or +65/-125 milliseconds, whichever is greater, not including instrumentation errors, of the set time at anbient temperature and +2.5%/-2.0% or +65/-125 milliseconds, whichever is greater, not including instrumentation errors, of the set time at hot and cold temperatures. The firing output pulse shall provide at least 1,000 ergs in less than 100 microseconds into an M84 detonator or equivalent load.

3.15 Module thermal shock. The module shall withstand the thermal shock test per MIL-STD-331, Test C7.

3.16 S&A Position. The safe position of the S&A's shall be verified just prior to issuance for fuze final assembly; limited to a maximum of a half day's production. After insertion of the S&A, fuze handling shall be controlled to preclude any chance of inadvertent arming.

3.17 Charge circuit continuity (fuze unarmed). The unarmed fuze shall be tested for continuity through its umbilical lead with a government approved test set. If there is no government approved test set available, the contractor shall design a charge circuit continuity test set that provides an output pulse of 40 mamps maximum for a period of 0.10 msec. maximum.

NOTE: This electrical characteristic is an electrical interface characteristic and shall not be altered or changed without an approved Government Deviation/Waiver or ECP.

The accept/reject requirement of the umbilical cable and the charge circuit of the firing capacitor is:

a. Reject as an open circuit if measured resistance is greater than the resistance of a 2.25 kohm resistance connected in series with the charge circuit of the firing capacitor.

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b. Reject as a short if the measured resistance is less than the resistance of a 220 ohm resistance connected in parallel with the charge circuit of the firing capacitor.

The test set shall be calibrated with acceptable production modules using the above modifications. Thirty fuzes shall be modified for the short condition and thirty fuzes shall be modified for the open condition. The results of the calibration process shall be subject to government review and approval.

3.18 Environmental conditioning. The fuze shall meet all performance requirements after being subject to the following environmental conditions.

3.18.1 Temperature and humidity. Temperature and humidity shall be in accordance with MIL-STD-331, Test C1, limited to one 14 day cycle.

3.18.2 Thermal shock. Thermal Shock shall be in accordance with MIL-STD-331, Test C7.

3.19 Safety.

3.19.1 Jolt and jumble. The fuze shall be subjected to and shall withstand the Jolt Test as specified in MIL-STD-331 Test A1, Jolt Machine Method, with the exception that the fuze shall not be disassembled and inspected after the Jolt Test. After completion of the Jolt Test, the same fuze shall be subjected to and shall withstand the Jumble Test as specified in MIL-STD-331, Test A2.

3.19.2 12 meter (40 foot) drop. The complete fuze, warhead P/N 9362895 and inert motor P/N 233AS431 assembly shall withstand the 12 Meter Drop Test as specified in MIL-STD-331, Test A3. To satisfy the requirements of MIL-STD-331, the disassembled fuze must be safe to handle (see 6.7.1).

3.19.3 One and one-half meter (5 foot) drop. The fuze shall be safe to handle, fire, or dispose of following the One and One-Half Meter Drop Test as specified in MIL-STD-331, Test A4.1, Single Drop Procedure. It needs not be operable.

3.19.4 Non-arming. The fuze shall not arm when subjected to an arming acceleration of 12.0 ± 0.5 g for a minimum of 5 seconds.

3.19.5 Mechanical interrupt. The fuze shall be incapable of being charged in the armed condition.

3.20 Fuze operational. The fuzes shall be subjected to and shall withstand the following operational tests: Non-Environmental Test, Helicopter Vibration Test (See Figure 1), Ballistic Test.

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3.21 Fuze arming. The fuze, after being temperature conditioned, shall meet the following requirements.

3.21.1 Ambient temperature arming time. The fuze assembly, after conditioning at a temperature of 70 ± 15°F for 2 hours minimum, shall arm between 0.70 and 0.90 second when subjected to an arming acceleration of 55.0 ± 0.5 g.

3.21.2 Low temperature arming time. The fuze after being conditioned at a temperature of -50° ± 5°F for 8 hours minimum, shall arm between 0.70 and 0.90 second when subjected to an arming acceleration of 55.0 ± 0.5 g.

3.21.3 High temperature arming time. The fuze after being conditioned at a temperature of +150°F ± 5°F for 8 hours minimum shall arm between O.70 and 0.90 second when subjected to an arming acceleration of 55.0 ± 0.5 g.

3.22 Charge-arm-function. The fuze shall be charged with a government approved. Ballistic Firing Test Set through its umbilical terminals, armed in accordance with fuze requirement 3.21.1 and shall, upon the closure of its setback weight switch, exhibit corresponding function time delays as shown in 4.5.35.2.

3.23 Ballistic firing. The fuze shall perform properly following temperature conditioning for a minimum of 8 hours after stabilization at: -50°F ± 5°F, 70°F ± 15°F (ambient) and +150°F ±5°F. The fuze shall function in accordance with 4 .5.35.2 after motor ignition. When properly charged and armed, the fuze shall function the detonator high order on expiration of the preset time delay.

3.24 Data requirements. The contractor shall generate data in accordance with the requirements of the data item descriptions cited in 6.3.

3.25. Workmanship.

3.25.1 Mechanical. All parts and assemblies shall be free from burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, porosity, warpage, burn marks, checks, blisters, excess flash, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agents. Surface coating shall be continuous except for a few slight scratches not exposing base materials. All required markings shall be neat and legible.

3.25.2 Electrical. Cable assemblies and wires shall be dressed in such manner as to prevent pinching or chafing of the wire Insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust, or foreign material.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as a part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 General provisions Unless otherwise specified herein, the previsions of MIL-A-48078 apply and form a part of this specification. Reference shall be made to MIL-STD-109 to define quali ty assurance terms used herein.

4.2 Classification of inspections. The following types of inspection requirements specified 'herein are classified as follows:

a. First article inspection (see 4.3) .

b. Quality conformance inspection (see 4.4).

4.3 First article inspection.

4.3.1 Submission, The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the assemblies, components and test specimens Listed below in the quantities indicated.

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Part. Description	Drawing	Quantity
Spring, Detent	8883713	10
Spring, Setback Weight	8883718	20
Strap, Rotor Ground	9260798	20
Speed Clip	9260891	20
stud, Spring	8883721	10
Rotor Housing	9260795	10
Socket, Detonator	9260799	25
Ground Contact, S&A	9352734	25
Contact, Common	9260783	25
Contact, N.C.	9260785	25
Contact, Detonator	9260787	25
Contact, Ground	9260788	25
Contact, N.O.	9260784	25
Plate, Top	9260781	25 or 5 per cavity (whichever is greater)
Plate, Bottom	9260782	25 or 5 per cavity (whichever is greater)
Gear and. Pinion Assembly No. 1	8883703	25
Gear and Pinion Assembly No. 2	8883700	25
Escapement Gear and Pinion Assembly	8883710	25
Lever Assembly	9260776	30
Rotor Assembly	9260796	20
Rotor Housing Assembly	9260794	10
Inner Plate Assembly	9260804	20
Escapement Assembly	9260774	20
Setback weight Assembly	9260790	25
Bottom Plate Assembly	9260780	25
S/A Assembly w/o switch	9260779	25
S/A Assembly (M439 Rocket Fuze)	9260778	45
S/A Housing	9260699	25 or 5 per cavity (which ever is greater)
Pin, interfacing	9260697-1	10
Pin, interfacing	9260697-2	10
Electronic Assembly	9260689	25
Molded Electronics Assembly (mold)	9260700	25 or 5 per cavity (whichever is greater)
Molded Electronics Assembly (Module)	9260700	486
Umbilical Ground Pin	9260719	25
Umbilical Charge Pin	9260720	25
Electronics Housing	9260698	25 or 5 per cavity (whichever is greater)
Electronic Housing Assembly	9260721	25
Fuze Rocket M439	9260704	246

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TABLE II. First article inspection.

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Fuze Rocket, M439	1 OF 9	See Below
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Spring, Detent			
	(Dwg. 8883713)			
	Examination for Defects	10	3.4	4.4.2.1
	Spring Embrittlement	10	3.5	4.5.3
	Spring, Setback Weight			
	(Dwg. 8883718)			
	Examination for Defects	20	3.4	4.4.2.2
	Spring Embrittlement	10	3.5	4.5.3
	Setback Weight Spring Set	10	3.6	4.5.4
	Strap, Rotor Ground			
	(Dwg. 9260798)			
	Examination for Defects	20	3.4	4.4.2.3
	Speed Clip			
	(Dwg. 9260891)			
	Examination for Defects	20	3.4	4.4.2.4
	Stud, Spring			
	(Dwg. 8883721)			
	Examination for Defects	10	3.4	4.4.2.5

NOTES

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TABLE II. First article inspection.

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE Fuze Rocket, M439	SHEET 2 OF 9	DRAWING NUMBER See Below
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY / INSPECTION METHOD REFERENCE
	Rotor Housing (Dwg. 9260795) Examination for Defects	10	3.4	4.4.2.6
	Socket, Detonator (Dwg. 9260799) Examination for Defects	25	3.4	4.4.2.7
	Ground Contact, S&A (Dwg. 9352734) Examination for Defects	25	3.4	4.4.2.8
	Contact, Common (Dwg. 9260783) Examination for Defects	25	3.4	4.4.2.9
	Contact, N.C. (Dwg. 9260785) Examination for Defects	25	3.4	4.4.2.10

NOTES.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE Fuze Rocket, M439	SHEET 3 OF 9	DRAWING NUMBER See Below
			NEXT HIGHER ASSEMBLY
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Contact, Detonator (Dwg. 9260787) Examination for Defects	25	3.4	4.4.2.11
	Contact, Ground (Dwg. 9260788) Examination for Defects	25	3.4	4.4.2.12
	Contact, N.O. (Dwg. 9260784) Examination for Defects	25	3.4	4.4.2.13
	Plate, Top (Dwg. 9260781) Examination for Defects	25(a)	3.4	4.4.2.14
	Plate, Bottom (Dwg. 9260782) Examination for Defects	25(a)	3.4	4.4.2.15

NOTES

a. Minimum of 5 per cavity for multiple mold with at least 25 total.

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TABLE II. First article inspection.

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE Fuze Rocket, M439	SHEET 4 OF 9	DRAWING NUMBER See Below
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY INSPECTION METHOD REFERENCE
	Gear and Pinion Assembly No. 1 (Dwg. 8883703)			
	Examination for Defects	25	3.4	4.4.2.16
	Load Test of No. 1 Gear and Pinion Assembly	25	3.4	4.5.6.1
	Gear and Pinion Assembly No. 2 (Dwg. 8883700)			
	Examination for Defects	25	3.4	4.4.2.17
	Load Test of No. 2 Gear and Pinion Assembly	25	3.4	4.5.6.2
	Escapement Gear and Pinion Assembly (Dwg. 8883710)			
	Examination for Defects	25	3.4	4.4.2.18
	Load Test of Escapement Gear and Pinion Assembly	25	3.4	4.5.6.3
	Lever Assembly (Dwg. 9260776)			
	Examination for Defects	30	3.4	4.4.2.19
	Load Test of Lever Shaft	10	3.4	4.5.6.4
	Load Test of Lever Pallet	10	3.4	4.5.6.5
	Torque Test of Lever Shaft	10	3.4	4.5.6.6

NOTES:

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TABLE II. First article inspection.

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE Fuze Rocket, M439	SHEET 5 OF 9	DRAWING NUMBER See Below
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Rotor Assembly			
	(Dwg. 9260796)			
	Examination for Defects	20	3.4	4.4.2.20
	Load Test of Annular Gear	10	3.4	4.5.6.7
	Load Test of Lock Roller Pin	10	3.4	4.5.6.8
	Rotor Housing Assembly			
	(Dwg. 9260794)			
	Examination for Defects	10	3.4	4.4.2.21
	Load Test of Spring Stud	10	3.4	4.5.6.9
	Inner Plate Assembly			
	(Dwg. 9260774)			
	Examination for Defects	20	3.4	4.4.2.22
	Load Test of Bridge Rivet	20	3.4	4.5.6.10
	Escapement Assembly			
	(Dwg. 9260774)			
	Examination for Defects	20	3.4	4.4.2.23
	Load Test of Pillars	20	3.4	4.5.6.11

NOTES

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TABLE II. First article inspection.

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Fuze Rocket, M439	6 OF 9	See Below
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Setback Weight Assembly			
	(Dwg. 9260790)			
	Examination for Defects	25	3.4	4.4.2.24
	Pull Off Test of Insulator	25	3.4	4.5.7
	Pull Off Test of Switch Pin	25	3.4	4.5.8
	Bottom Plate Assembly			
	(Dwg. 9260780)			
	Examination for Defects	25	3.4	4.4.2.25
	S&A Assembly w/o Switch			
	(Dwg. 9260779)			
	Examination of Defects	25	3.4	4.4.2.26
	Setback Weight Return Capability	25	3.7	4.5.9
	Torque Test of Rotor Stop Stud	25	3.4	4.4.5.11
	S&A Assembly (M439 Rocket Fuze)			
	(Dwg. 9260778)			
	Examination for Defects	45	3.4	4.4.2.27
	Rotor Reset Capability	25	3.8	4.4.5.10
	Load Test of Detonator	45	3.13.1	4.5.12
	Continuity through Detonator	45	3.13.2	4.5.13
	Non-Arming (a)	45	3.13.3	4.5.14, 4.5.15
	S&A Arming Time (a)	45	3.13.4	4.5.14, 4.5.16

NOTES.

a. This test may be performed with a simulated detonator.

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TABLE II. First article inspection.

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		SHEET 7 OF 9	DRAWING NUMBER
	Fuze Rocket, M439			See Below
				NEXT HIGHER ASSEMBLY
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	S/A Housing			
	(Dwg. 9260699)			
	Examination for Defects	25(c)	3.4	4.4.2.28
	Pin Interfacing			
	(Dwg. 9260697)			
	Examination for Defects	10(b)	3.4	4.4.2.29
	Electronic Assembly			
	(Dwg. 9260689)			
	Examination for Defects	25	3.4	4.4.2.30
	Module Time Delay	25	3.14	4.5.17(a)
	Molded Electronics Assembly			
	(Dwg. 9260700)			
	Examination for Defects	25(c)	3.4 and 3.1.1	4.4.2.31
	Module Time Delay	486	3.14	4.5.17/Table VI
	Thermal Shock	36	3.15	4.5.18/Table VI

NOTES:
a. Perform at ambient temperature only.
b. Quantity of 10 for each type required.
c. Minimum 5 per cavity for multiple mold with at least 25 total.

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TABLE II. First article inspection.

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Fuze Rocket, M439	8 OF 9	See Below
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Umbilical Charge Pin (Dwg. 9260719) Examination for Defects	25	3.4	4.4.2.32
	Umbilical Charge Pin (Dwg. 9260720) Examination for Defects	25	3.4	4.4.2.33
	Electronics Housing (Dwg. 9260698) Examination for Defects	25(a)	3.4 and 3.1.1	4.4.2.34
	Electronics Housing Assembly (Dwg. 9260721) Examination for Defects	25	3.4	4.4.2.35

NOTES

a. Minimum of 5 per cavity for multiple mold with at least 25 total.

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TABLE II. First article inspection.

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Fuze Rocket, M439	9 OF 9	See Below
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Fuze Rocket, M439 (Dwg. 9260704)			
	Examination for Defects	246	3.4	4.4.2.36
	S&A Position Check	246	3.16	4.5.19
	Charge Circuit Continuity	246	3.17	4.5.20
	Temperature and Humidity	20	3.18.1	4.5.21
	Thermal Shock	36	3.18.2	4.5.22
	Jolt	12	3.19.1	4.5.23
	Jumble	12	3.19.1	4.5.23
	Forty Foot Drop	15	3.19.2	4.5.24
	Five Foot Drop	15	3.19.3	4.5.25
	Non-Environmental Control	68	3.20	4.5.26
	Helicopter Vibration	80	3.20	4.5.27
	Charge Circuit Continuity (Following Helicopter Vibration)	80	3.17	4.5.20
	Non-Arm	32	3.19.4	4.5.29
	Arming Time (Under Minimum)	32	3.21	4.5.30
	Arming Time (Over Max. or Fail to Arm)	32	3.21	4.5.30
	Mechanical Interrupt Test	32	3.19.5	4.5.31
	Ballistic Firing	120	3.23	4.5.35

NOTES

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4.3.2 Inspections to be performed. As determined by the Government, the first article assemblies, components, and test specimens may be subjected to any or all of the examinations and tests specified in this detail specification and be inspected for compliance with any or all requirements of the applicable drawings. See Table II for inspection and testing to be performed.

4.3.3 Rejection. If any assembly, component, or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of an assembly, component, or test specimen to comply with any of the requirements.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term “inspection lot” is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the Inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of the M439 Fuze shall contain the following:

a. Deter.ators from not more than three detonator lots with the same interfix number from one manufacturer.

b. Metal parts bearing the same metal parts lot interfix number from one metal parts manufacturer.

4.4.2 Examination and tests.

a. Classification of characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contract's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified. When cited herein, attributes sampling inspection shall be conducted in accordance with Table III below, using the inspection levels stated in the Classification of Characteristics paragraphs.

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TABLE III. Attributes sample inspection.

Lot Size	Inspection Levels					
	S-4	II	III	IV	V	VI
2 to 8	*	*	*	*	*	5
9 to 15	*	*	*	*	13	5
16 to 25	*	*	*	20	13	5
26 to 50	32	*	32	20	13	5
51 to 90	32	*	32	20	13	7
91 to 150	32	125	32	20	13	11
151 to 280	32	125	32	20	20	13
281 to 500	32	125	48	47	29	16
501 to 1200	32	125	73	47	34	19
1201 to 3200	32	125	73	53	42	23
3201 to 10000	32	192	86	68	50	29
10001 to 35000	32	294	108	77	60	35
35001 to 150000	73	294	123	96	74	40
150001 to 500000	1250	500	200	125	50	13
500001 and over	1250	500	200	125	50	13

Numbers under inspection levels indicate sample size; asterisks (*) indicate one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection. Accept on zero and reject on one or more for all inspection levels.

b. Alternative quality conformance provisions. Unless otherwise specified herein or provided for in the contract, alternative conformance procedures, methods or such as statistical process control, tool control, other types of sampling procedures, etc. may be used be the contractor when they provide as a minimum, the level of quality assurance required by the provisions cited herein. Prior to applying such alternative procedures, methods or equipment, the contractor shall, describe them in a written proposal submitted to the Government for evaluation (see 6.6). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality assurance provisions herein. In cases of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program or detailed inspection system, as applicable.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Spring, Detent			8883713
				NEXT HIGHER ASSEMBLY
				9260774
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Spring embrittlement	10	3.5	4.5.3
Minor				
201	Evidence of poor workmanship	Level IV	3.25	Visual

NOTES

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Spring, Setback Weight	1 OF 1	8883718
			NEXT HIGHER ASSEMBLY: 9260774

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Spring embrittlement	10	3.5	4.5.3
102	Setback weight spring set	10	3.6	4.5.4
Minor				
201	Evidence of poor workmanship	Level IV	3.25	Visual

NOTES

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-F-48877A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Strap, Rotor Ground	1 OF 1	9260798
			NEXT HIGHER ASSEMBLY: 9260796

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Width at detonator end	Level III	3.4	Gage
102	Strap improperly bent	Level III	3.4	Gage/Visual
103	Location of detonator end	Level III	3.4	Gage
Minor				
201	Evidence of poor workmanship	Level IV	3.25	Visual

NOTES

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CLASSIFICATION OF CHARACTERISTICS

MIL-F-48877A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Speed Clip	1 OF 1	9260891
			NEXT HIGHER ASSEMBLY: 9260778

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Length of clip	Level III	3.4	Gage
102	Diameter of clip	Level III	3.4	Gage
Minor				
201	Evidence of poor workmanship	Level IV	3.25	Visual

NOTES

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CLASSIFICATION OF CHARACTERISTICS

MIL-F-48877A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Stud, Spring	1 of 1	8883721
			NEXT HIGHER ASSEMBLY: 9260794

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Diameter of stud, min.	Level III	3.4	Gage
Minor				
01	Evidence of poor workmanship	Level IV	3.25	Visual

NOTES

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CLASSIFICATION OF CHARACTERISTICS

MIL-F-48877A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Rotor Housing	1 of 1	9260795
			NEXT HIGHER ASSEMBLY: 9260794

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Diameter of base	Level III	3.4	Gage
102	Rotor housing post separation	Level III	3.4	Gage
103	True position of eyelet holes, 2 places	Level III	3.4	Gage
104	Diameter of eyelet holes, 2 places	Level III	3.4	Gage
Minor				
201	Evidence of poor workmanship	Level IV	3.25	Visual

NOTES

AMSMC Form 1570a-E, 1 Jul 89 Replaces AMSMC Form 1570a, 1 Apr 85, which may not be used.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7	Socket, Detonator	1 OF 1	9260799
			NEXT HIGHER ASSEMBLY 9260796

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Detonator pin socket diameter	Level III	3.4	Gage
Minor				
201	Evidence of poor workmanship	Level IV	3.25	Visual

NOTES

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.8	Ground Contact, S&A	1 OF 1	9352734
			NEXT HIGHER ASSEMBLY: 9260780

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Radius of bend (70 deg + 5 deg)	Level III	3.4	Gage
102	Diameter of eyelet hole	Level III	3.4	Gage
103	Thickness of strip	Level III	3.4	Gage
104	Length of contact arm	Level III	3.4	Gage
105	Poor quality of plating	Level III	3.4	Visual
Minor				
201	Evidence of poor workmanship	Level IV	3.25	Visual

NOTES

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CLASSIFICATION OF CHARACTERISTICS

MIL-F-48877A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	ontact, Common	1 of 1	9260783
			NEXT HIGHER ASSEMBLY: 9260780

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Angle of bend (82 deg + 6 deg)	Level III	3.4	Gage
102	Angle of bend (61 deg + 6 deg)	Level III	3.4	Gage
103	True position of eyelet hole	Level III	3.4	Gage
104	Diameter of eyelet hole	Level III	3.4	Gage
105	Thickness of strip	Level III	3.4	Gage
106	Poor quality of plating	Level III	3.4	Visual
107	Angle of bend (64 deg + 6 deg)	Level III	3.4	Gage
Minor				
201	Evidence of poor workmanship	Level IV	3.25	Visual

NOTES

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CLASSIFICATION OF CHARACTERISTICS

MIL-F-48877A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.10	Contact, N.C.	1 OF 1	9260785
			NEXT HIGHER ASSEMBLY: 9260780

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Diameter of eyelet hole	Level III	3.4	Gage
102	Angle of bend (86 deg + 6 deg F)	Level III	3.4	Gage
103	Thickness of strip	Level III	3.4	Gage
104	Poor quality of plating	Level III	3.4	Visual
105	Angle of bend (64 deg + 6 deg)	Level III	3.4	Gage
Minor				
201	Evidence of poor workmanship	Level IV	3.25	Visual

NOTES

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CLASSIFICATION OF CHARACTERISTICS

MIL-F-48877A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4...11	Contact, Detonator	1 OF 1	9260787
			NEXT HIGHER ASSEMBLY: 9260780

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
201	Angle of bend (45 deg + 6 deg)	Level III	3.4	Gage
202	Angle of bend (86 deg + 6 deg)	Level III	3.4	Gage
203	Diameter of eyelet hole	Level III	3.4	Gage
204	Thickness of strip	Level III	3.4	Gage
205	Poor quality of plating	Level III	3.4	Visual
206	Angle of bend (64 deg + 6 deg)	Level III	3.4	Gage
Minor				
301	Evidence of poor workmanship	Level IV	3.25	Visual

NOTES

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.7.12	Contact, Ground	1 of 1	9260788
			NEXT HIGHER ASSEMBLY: 9260780

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Angle of bend (90 deg + 4 deg)	Level III	3.4	Gage
102	True position of eyelet hole	Level III	3.4	Gage
103	Diameter of eyelet hole	Level III	3.4	Gage
104	Thickness of strip	Level III	3.4	Gage
105	Poor quality of plating	Level III	3.4	Visual
106	Angle of bend (64 deg + 6 deg) 2 places	Level III	3.4	Gage
Minor				
201	Evidence of poor workmanship	Level IV	3.25	Visual

NOTES

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CLASSIFICATION OF CHARACTERISTICS

MIL-F-48877A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.13	Contact, N.O.	1 of 1	9260784
			NEXT HIGHER ASSEMBLY: 9260780

CLASSIFICATION		EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical		None defined.			
Major					
	01	True position of eyelet hole	Level III	3.4	Gage
	02	Diameter of eyelet hole	Level III	3.4	Gage
	03	True position of dimple locator hole	Level III	3.4	Gage
	04	Angle of bend (87 deg ± 6 deg)	Level III	3.4	Gage
	05	Thickness of strip	Level III	3.4	Gage
	06	Poor quality of plating	Level III	3.4	Visual
	07	Angle of bend (64 deg ± 6 deg)	Level III	3.4	Gage
Minor					
	01	Evidence of poor workmanship	Level IV	3.25	Visual

NOTES

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.14	Plate Top	1 OF 1	9260781
			NEXT HIGHER ASSEMBLY: 9260778

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Mold inspection	5/cavity	3.4	4.5.2
102	True position of eyelet holes, 2 places	Level III	3.4	Gage
103	Diameter of eyelet holes, 2 places	Level III	3.4	Gage
104	True position of eyelet bosses, 2 places	Level III	3.4	Gage
105	True position of S&A ground contact hole	Level III	3.4	Gage
106	True position of contact locator holes, 4 places	Level III	3.4	Gage
107	True position of contact locator holes, 3 places	Level III	3.4	Gage
108	Location of common contact post	Level III	3.4	Gage
109	Outside diameter	Level III	3.4	Gage
Minor				
201	Evidence of poor workmanship	Level IV	3.25	Visual

NOTES

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
1.4.2.15	Plate, Bottom	1 OF 1	9260782

NEXT HIGHER ASSEMBLY: 9260780

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Mold inspection	5/cavity	3.4	4.5.2
102	True position of eyelet holes, 2 places	Level III	3.4	Gage
103	Diameter of eyelet holes, 2 places	Level III	3.4	Gage
104	True position of contact eyelet holes, 5 places	Level III	3.4	Gage
105	Diameter of contact dimples, 6 places	Level III	3.4	Gage
106	True position of contact dimples, 6 places	Level III	3.4	Gage
107	Outside diameter	Level III	3.4	Gage
Minor				
201	Evidence of poor workmanship	Level IV	3.25	Visual

NOTES

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CLASSIFICATION OF CHARACTERISTICS

MIL-F-48877A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.16	Gear and Pinion Assembly No. 1	1 of 1	8883703
			NEXT HIGHER ASSEMBLY: 9260774

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical				
1	Maximum gap between gear and pinion	100%	3.4	Gage
2	Load test of No. 1 gear and pinion assembly	100%	3.4	4.5.6.1
Major	None defined			
Minor				
201	Evidence of poor workmanship	Level IV	3.25	Visual

NOTES

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CLASSIFICATION OF CHARACTERISTICS

MIL-F-48877A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.17	Gear and Pinion Assembly No. 2	1 OF 1	8883700
			NEXT HIGHER ASSEMBLY: 9260774

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical				
1	Maximum gap between gear and pinion	100%	3.4	Gage
2	Load test of No. 2 gear and pinion assembly	100%	3.4	4.5.6.2
Major	None defined			
Minor				
201	Evidence of poor workmanship	Level IV	3.25	Visual

NOTES

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.18	Escapement Gear and Pinion Assembly	1 OF 1	8883710
			NEXT HIGHER ASSEMBLY: 9260774

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical				
1	Maximum gap between gear and pinion	100%	3.4	Gage
2	Load test of escapement gear and pinion assembly	100%	3.4	4.5.6.3
Major	None defined			
Minor				
201	Evidence of poor workmanship	Level IV	3.25	Visual

NOTES

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.19	Lever Assembly	1 OF 1	9260776
			NEXT HIGHER ASSEMBLY: 9260774

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical				
1	Maximum gap between lever and shaft	100%	3.4	Gage
2	Load test of lever shaft	100%	3.4	4.5.6.4
3	Load test of lever pallet	100%	3.4	4.5.6.5
Major				
101	Torque test of lever shaft	100%	3.4	4.5.6.6
Minor				
201	Evidence of poor workmanship	Level IV	3.25	Visual

NOTES

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.10	Rotor Assembly		9260796
			NEXT HIGHER ASSEMBLY 9260779

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical				
1	Load test of annular gear	100%	3.4	4.5.6.7
Major				
101	Load test of lock roller pin	Level III	3.4	4.5.6.8
102	Banking stud pullout test	Level III	3.4	Gage
103	Rotor ground strap and detonator socket contact resistance	Level III	3.4	Milliohmmeter
Minor				
201	Evidence of poor workmanship	Level IV	3.25	Visual

NOTES

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CLASSIFICATION OF CHARACTERISTICS

MIL-F-48877A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.21	Rotor Housing Assembly	1 OF 1	9260794
			NEXT HIGHER ASSEMBLY: 9260779

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
201	Load test of spring stud	Level III	3.4	4.5.6.9
Minor				
101	Evidence of poor workmanship	Level IV	3.25	Visual

NOTES

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.22	Inner Plate Assembly	1 OF 1	9260804
			NEXT HIGHER ASSEMBLY: 9260774

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Load test of bridge rivet	Level III	3.4	4.5.6.10
Minor				
201	Evidence of poor workmanship	Level IV	3.25	Visual

NOTES

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.23	Escapement Assembly	SHEET 1 OF 1	9260774
			NEXT HIGHER ASSEMBLY
			9260779

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Load test of pillars	Level III	3.4	4.5.6.11
Major				
201	Evidence of poor workmanship	Level IV	3.25	Visual

NOTES

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.24	Setback Weight Assembly	1 OF 1	9260790
			NEXT HIGHER ASSEMBLY: 9260779

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Switch pin insulator - missing, scored, with voids or cracks	Level III	3.4	Visual
102	Perpendicularity of Switch Pin	Level III	3.4	Gage
103	Maximum gap between insulation and side of "G" weight	Level III	3.4	Gage
104	Pull-off test of insulator (a)	Level S-4	3.4	4.5.7
105	Pull-off test of Switch Pin (a)	Level S-4	3.4	4.5.8
Minor				
201	Evidence of poor workmanship	Level IV	3.25	Visual

NOTES

a. The Pull-off Test of the insulator shall be performed before the Pull-off Test of the Switch Pin. The same sample units shall be used for each test.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-F-48877A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.25	Bottom Plate Assembly	1 OF 1	9260780
			NEXT HIGHER ASSEMBLY: 9260778

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
01	Eyelets above flush, 5 places	Level III	3.4	Visual
02	Alignment of N.C. contact	Level III	3.4	Visual
03	Alignment of common contact	Level III	3.4	Visual
04	Alignment of N.O. contact	Level III	3.4	Visual
05	Alignment of ground contacts	Level III	3.4	Gage
Minor				
01	Evidence of poor workmanship	Level IV	3.25	Visual

NOTES

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.26	S&A Assembly w/o Switch	SHEET 1 OF 2	9260779
			NEXT HIGHER ASSEMBLY: 9260778

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical				
1	Distance between annular gear and No. 2, gear, max.	100%	3.9	Gage
2	Setback weight return capability	100%	3.7	4.5.9
Major				
101	Detonator socket contact missing	Level III	3.4	Visual
102	Rotor ground strap missing	Level III	3.4	Visual
103	Rotor ground strap deformed	Level III	3.4	Visual
104	Rotor ground strap not making contact with detonator socket contact	100%	3.4	Visual
105	Foreign matter present	Level III	3.4	Visual
106	Lubricant missing	Level III	3.4	Visual
107	Diameter across outside corners of posts of rotor housing min.	Level III	3.4	Gage
108	Protective finish missing or incomplete	Level III	3.4	Visual

NOTES

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-F-48877A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.26	S&A Assembly w/o Switch	2 OF 2	9260779
			NEXT HIGHER ASSEMBLY 9260778

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Major				
109	Rotor Assembly does not lock in fully armed position	Level III	3.4	Visual
110	Alignment of detonator cavity with the outside diameter	Level III	3.10	Gage
111	Torque test of rotor stop stud	Level III	3.4	4.5.11
112	Wrong lubricant	100%	3.4	Certification
113	Excessive lubricant (a)	Level III	3.4	Visual
Minor				
201	Evidence of poor workmanship	Level IV	3.25	Visual

NOTES

a. Excessive lubricant is defined as evidence of non-uniform coating or collection of fluid in visible areas.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-F-48877A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.27	S&A Assembly (M439 Rocket Fuze)	1 of 2	9260778
			NEXT HIGHER ASSEMBLY: 9260704

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical				
1	Non-arming (a)	100%	3.13.3	4.5.15
2	Rotor reset capability	100%	3.8	4.5.10
Major				
101	Overall length, max./min.	Level III	3.4	Gage
102	S&A improperly assembled and seated in top plate	Level III	3.4	Visual
103	Contact resistance between No. 1 and and No. 5 contacts (safe)	100%	3.11	Milliohmmeter
104	Insulation resistance of No. 1 and No. 2 contacts (safe)	100%	3.11	Megohmmeter, test voltage 500 VDC
105	Insulation resistance of No. 1 and No. 5 contacts (armed w/o detonator)	100%	3.11	Megohmmeter, test voltage 500 VDC
106	Contact resistance between No. 1 and No. 2 contacts (armed w/o detonator)	100%	3.11	Milliohmmeter
107	Contact resistance between No. 4 and No. 6 contacts and rotor (armed w/o detonator)	100%	3.11	Milliohmmeter

NOTES

a. This shall be last test prior to insertion of the S&A Assy into the fuze Dwg. 9260704.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 2 OF 2	DRAWING NUMBER
4.4.2.27	S&A Assembly (M439 Rocket Fuze)		9260778
			NEXT HIGHER ASSEMBLY 9260704

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
108	Speed clip protrudes beyond outer periphery of rotor after assembly or less than two speed clip tabs retaining detonator	Level III	3.4	Visual
109	Rotor ground strap not contacting detonator socket terminal	Level III	3.4	Visual
110	Detonator perforated or damaged (c)	Level III	3.4	Visual
111	Load test of Detonator	Level III	3.13.1	4.5.12
112	Continuity through detonator (b)	100%	3.13.2	4.5.13
113	Arming Time (e)	100%	3.13.4	4.5.16 (d)
114	Eyelets too close to surface of bottom of plate assembly, 2 places	Level III	3.4	Gage
Minor				
201	Marking missing or incomplete	Level IV	3.4	Visual
202	Evidence of poor workmanship	Level IV	3.25	Visual

NOTES

b. This test shall consist of arming S&A, measuring continuity, returning rotor to safe position.
c. Detonators are to be tested in accordance with paragraph 4.5.5.
d. Perform at ambient temperature only.
e. May be performed with a simulated detonator.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.28	S&A Housing	1 OF 1	9260699
			NEXT HIGHER ASSEMBLY: 9260704

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Mold inspection	5/Cavity	3.4	4.5.2
102	Length from forward end of housing to aft step	Level III	3.4	Gage
103	Rotor outside diameter clearance radius	Level III	3.4	Gage
104	Inside diameter of housing	Level III	3.4	Gage
105	Outside diameter of housing flange	Level III	3.4	Gage
106	Depth of S&A cavity	Level III	3.4	Gage
107	Forward inside radius, 2 places	Level III	3.4	Gage
108	Width between rotor housing guides	Level III	3.4	Gage
109	Width of index slots	Level III	3.4	Gage
110	Thickness of index slots	Level III	3.4	Gage
111	Outside diameter of housing	Level III	3.4	Gage
112	Width between rotor housing guides	Level III	3.4	Gage
113	Thickness of rotor housing guides	Level III	3.4	Gage
Minor				
201	Evidence of poor workmanship/molding	Level IV	3.25/3.4	Visual

NOTES

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.29	Pin Interfacing (Applicable to Pins Only)	1 OF 1	9260697
			NEXT HIGHER ASSEMBLY: 9260689

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Longer length after bending	Level III	3.4	Gage
102	Shorter length after bending	Level III	3.4	Gage
103	Diameter of wire	Level III	3.4	Gage
104	Wire trip flat over max.	Level III	3.4	Gage
Minor				
201	Evidence of poor workmanship	Level IV	3.25	Visual

NOTES

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		SHEET 1 OF 1	DRAWING NUMBER
4.4.2.30	Electronic Assembly			9260689
				NEXT HIGHER ASSEMBLY
				9260700
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Defective soldered connections	Level III	3.4	Visual
102	Components improperly mounted on substrate	Level III	3.4	Visual
103	Components missing from substrate	Level III	3.4	Visual
104	Coating missing, incomplete or defective	Level III	3.4	Visual
Minor				
201	Evidence of poor workmanship	Level IV	3.25	Visual

NOTES

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.31	Molded Electronic Assembly	1 OF 2	9260700
			NEXT HIGHER ASSEMBLY: 9260704

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Mold inspection	5/Cavity	3.4	4.5.2
102	Length of interface pins from mold at center line of pins	Level III	3.4	Gage
103	True position of pins to reference diameter A	Level III	3.4	Gage
104	Outside diameter of mold, min.	Level III	3.4	Gage
105	Overall length	Level III	3.4	Gage
106	Function time delay: 2 seconds	100%	3.14	4.5.17 (a)
107	Function time delay: 10 seconds	100%	3.14	4.5.17 (a)
108	Function time delay: 24 seconds	100%	3.14	4.5.17 (a)
109	Thermal Shock	30	3.15	4.5.18 (a)

NOTES

a. Perform at ambient temperature only.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		SHEET 2 OF 2	DRAWING NUMBER
4.1.2.31	Molded Electronic Assembly			9260700
				NEXT HIGHER ASSEMBLY 9260704
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Minor				
201	Fins not clean of mold material or have improper spherical termination	Level IV	3.4	Visual
202	Mold cavity identification not permanent or illegible	Level IV	3.4	Visual
203	Excessive molding defects, or a component or substrate is exposed	Level IV	3.4	Visual
204	Evidence of poor workmanship	Level IV	3.25	Visual

NOTES

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.32	Umbilical Ground Pin	1 OF 1	9260719
			NEXT HIGHER ASSEMBLY: 9260721

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Diameter of interface connector	Level III	3.4	Gage
102	Outside diameter of umbilical connector pin	Level III	3.4	Gage
103	Interface tip flat over max.	Level III	3.4	Gage
104	True position of umbilical connector pin to interface connector pin	Level III	3.4	Gage
Minor				
201	Absence of pin chamfer	Level IV	3.4	Visual
202	Evidence of poor workmanship	Level IV	3.25	Visual

NOTES

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		SHEET	DRAWING NUMBER
4.4.2.33	Umbilical Charge Pin		1 of 1	9260720
				NEXT HIGHER ASSEMBLY 9260721

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Diameter of interface connector pin	Level III	3.4	Gage
102	Outside diameter of umbilical connector pin	Level III	3.4	Gage
103	Interface tip flat over max.	Level III	3.4	Gage
104	True position of umbilical connector pin to interface connector pin	Level III	3.4	Gage
Minor				
201	Evidence of poor workmanship	Level IV	3.25	Visual

NOTES

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.34	Electronics Housing	1 OF 1	9260698
			NEXT HIGHER ASSEMBLY: 9260721

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Mold inspection	5/Cavity	3.4	4.5.2
102	True position of charge pin hole diameter to diameter A (a)	Level III	3.4	Gage
103	True position of ground pin hole diameter to diameter A (a)	Level III	3.4	Gage
104	True position of seal slot inside diameter to diameter A	Level III	3.4	Gage
105	Inside diameter of cavity	Level III	3.4	Gage
106	Diameter of charge pin hole (a)	Level III	3.4	Gage
107	Diameter of ground pin hole (a)	Level III	3.4	Gage
108	True Position of blind hole	Level III	3.4	Gage
109	Diameter of blind hole	Level III	3.4	Gage
110	Depth of blind hole, min.	Level III	3.4	Gage
Minor				
201	Width of seal slot	Level IV	3.4	Gage
202	Evidence of poor workmanship	Level IV	3.25/3.4	Visual

NOTES

a. If pins are molded into housing, categories 106 and 107 do not apply; categories 102 and 103 refer to pins instead of pin holes.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.35	Electronics Housing Assembly	1 OF 1	9260721
			NEXT HIGHER ASSEMBLY: 9260704

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Protrusion of connectors above flange	Level III	3.4	Gage
Minor				
201	Evidence of poor workmanship	Level IV	3.25	Visual

NOTES

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS .

MIL-F-48877A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.36	Fuze, Rocket, M439	1 OF 2	9260704

NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical				
1	S&A position check	100%	3.16	4.5.19
2	Non-environmental control	36	3.20	4.5.26
3	Non-arm	32 (a)	3.19.4	4.5.29
4	Arming time (under minimum)	32 (a)	3.21	4.5.30
Special (d)				
1	Jolt	12	3.19.1	4.5.23
2	Jumble	12	3.19.1	4.5.23
3	12 meter (40 foot) drop	15 (b)	3.19.2	4.5.24
4	One and one-half meter (5 foot) drop	15 (c)	3.19.3	4.5.25
5	Helicopter Vibration	65	3.20	4.5.27
6	Mechanical interrupt	32 (a)	3.19.5	4.5.31
Major				
101	Length of fuze, max.	Level III	3.4	Gage
102	Diameter of fuze, max.	Level III	3.4	Gage
103	Charge circuit continuity	100%	3.17	4.5.20

NOTES.

a. Samples taken from Helicopter vibration and Non-environmental.
b. First article only.
c. Five foot drop not done on every lot.
d. See MIL-A-48078 for definition.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		SHEET 2 OF 2	DRAWING NUMBER
4.4.2.36	Fuze, Rocket, M439			9260704
				NEXT HIGHER ASSEMBLY
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Major	(Continued)			
104	Arming time (over max. or fail to arm)	32 (a)	3.21	4.5.30
105	Temperature and Humidity	20 (b)	3.18.1	4.5.21
106	Charge circuit continuity (following helicopter vibration)	65	3.17	4.5.20
107	Thermal Shock	36 (b)	3.18.2	4.5.22
108	Ballistic firing	64 (a)	3.23	4.5.35
109	Module time delay (after centrifuge tear-down)	32	3.14	4.5.17.1
Minor				
201	Stamping missing	Level IV	3.4	Visual
202	Evidence of poor workmanship	Level IV	3.25	Visual

NOTES

a. Samples taken from Helicopter vibration and Non-environmental.
b. First article only.

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4.4.3 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See section 6.2 of MIL-A-48078 and section 6.4 herein. The contractor shall submit all inspection equipment designs to the government for evaluation and approval prior to fabrication of such inspection equipment.

4.4.4 Contractor inspection system. As part of contractor's approved quality procedure, there shall be a provision for checking the adequacy of inspections made in connection with listed critical or major defects. These procedures shall include the following provisions:

4.4.4.1 Handling procedure. A Design Agency approved procedure shall be instituted that shall preclude unauthorized handling of the S&A's and fuzes from the S&A position check until the S&A has been installed in the fuze. The procedure shall be able to eliminate armed fuzes from being introduced into the lot.

4.4.4.1.2 Identification of rejected material. All material listed below, rejected as containing defects, shall be isolated to preclude its inclusion with acceptable material:

a. All S&A's failing 4.5.9, 4.5.10, 4.5.15, 4.5.16, or 4.5.19.

b. All modules failing 4.5.17, or 4.5.18.

c. All fuzes failing 4.5.23, 4.5.24, 4.5.25, 4,5.27, or 4.5.28.

4.4.4.2 Use of simulated defect standards. The following listed verified Inspections shall be validated and inspection equipment verified once per shift using Design Agency approved item facsimiles or standards containing the defects and supplied by the contractor. The contractor shall resupply the item facsimiles when it is determined by the Government that the facsimiles are no longer representative enough of the product to make it unidentifiable to the contractor inspection personnel or it is not functioning in the necessary manner and cannot be corrected satisfactorily. These standards should be inert and readily identifiable to the Government to ensure recovery should they be inadvertently introduced into production. Simulated defect standards used to verify the following verified inspections should be constructed and identified in such a manner that they are not readily recognizable to the contractor Inspection personnel as defect standards. Standards should be

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used so that their number, time and sequence of introduction into the inspection cycle is known only to those personnel responsible for checking the accuracy of the operation. Adequacy of the standards shall be verified at least once each week.

Verified Inspection

Paragraph 4.5.9 - Setback Weight Return Capability.
Paragraph 4.5.19 - S&A Position Check.
Paragraph 4.5.17 - Module Time Delay Function Test.

NOTE: When there is automatic detection and automatic rejection of defective items to a controlled area or a controlled rejection of defective items, it is permissible for the defect standards to be identifiable and their introduction into production noticeable to contractor inspection personnel.

4.4.4.3 Undetected defect standards. Provisions shall be established for the corrective action to be taken where the equipment or contractor inspection personnel fail to identify a standard as defect. These provisions shall include a reinspection of all acceptable material subsequent to the last acceptable verification check as a minimum.

4.5 Methods of inspection. PRECAUTION: This specification covers sampling and testing of chemical, toxic or explosive materials which are potentially hazardous to personnel. According, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials.

4.5.3 Materials, components and processes . Compliance with all requirements of Section 3 of this specification shall be ascertained by current and continuing examination of inspection and test data to determine that all components (parts, subassemblies, and materials) have been inspected and tested and found to comply with their respective drawings and specification requirements, and that all specified manufacturing processes have been followed.

4.5.1.1 Visual and mechanical inspection. The visual, manual and mechanical inspections and tests shall verify compliance with the requirements of Sections 3 and 5 of this specification in accordance with 4.4.2 herein and Table II

4.5.2 Dimensional control of molded or plastic parts. In place of normal sampling and inspection associated with the Classification of Defects, and after a curing time for the parts has been approved (see Note 1) a sample of five (5) parts as molded from each cavity shall be fully inspected dimensionally to qualify a new or reworked cavity for use in production. The

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molded parts shall carry the individual cavity identification. As a control of each cavity during production, the above quantity of parts from each cavity shall be inspected after continuous production of each 5,000 parts or at the end of each week, whichever occurs first. Of the sample, one (1) part shall be the last produced. If any defective parts are found during qualification of the cavity, the cavity producing the defective part will not be used in production. If any defective parts are found when inspection is performed for the control of the cavity, the cavity producing the defective part shall be removed from production. Further, that portion of production since the last control check shall be returned to the contractor for inspection for each type of defect according to the level specified in the classification of defects. All cavities removed from production because of some fault, may after reworking be returned to production provided they pass the qualification test above. The contractor may request a change of inspection frequency provided he presents objective evidence to the contracting officer to substantiate the request.

NOTE 1: In establishing a curing time, dimensionally check ten (10) parts from each cavity at periodic intervals (e.g. every 30, 60, etc. mins.) until dimensional stability is attained. The curing time will be from the time the part comes out of the mold until dimensional stability is attained. The inspection data used in determining the curing time shall be available for inspection if required. If there is a change in material, or in the cycle time, or if a cavity is reworked, or a new cavity is used, a new curing time shall be established and approved.

NOTE 2: Mold samples may be returned to the lot after inspection.

4.5.3 Spring embrittlement. The spring shall be stretched along its longitudinal axis in the case of a coil spring, or bent in the case of a flat spring until the spring no longer returns to its original position. When the spring no longer returns to its original length or profile, it shall be considered distorted permanently. Any spring that breaks prior to this point shall be classed defective. One defect shall be cause for lot rejection.

4.5.4 Setback weight spring set. The spring shall be measured at its original free length, assembled in a restraining fixture in accordance with 4.4.3 and compressed to a normal height of .750 inches, placed in an oven and subjected to a temperature of not more than 180°F or less than 175°F for 24 hours. The spring shall be removed from the oven and cooled to ambient temperature. After cooling the spring for 24 hours

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minimum, the spring shall be measured and the measurement shall be compared to the original free length to ascertain compliance with the specified requirement. Any spring that fails the requirement shall be classed defective. One defect shall be cause for lot rejection.

4.5.5 Check test for deterioration of detonators. If the total time between original acceptance of any detonator lot and the assembly of that lot into the fuzes exceeds two years, at any time since previous tests, the detonator lot shall be subjected to and must satisfactorily pass the functioning and output tests specified in MIL-D-45495 immediately before the detonator lot is assembled into the fuzes. The test shall be performed by the contractor on detonators randomly selected by a Government Inspector. The test shall be conducted at Government expense without cost to the contractor who loaded the detonators or the contractor who is to assemble the detonators into the fuzes. The test shall not constitute a basis for fuze lot rejection against either contractor except where deterioration has occurred as a direct result of carelessness in handling, storage, etc. permitted while the detonator lot was under the jurisdiction of the contractor.

4.5.6 S&A mechanical tests.

4.5.6.1 Load test of No. 1 gear and pinion assembly. Apply the required axial load perpendicular to the No. 1 gear and in such a direction as to test the assembly of the No. 1 gear and pinion. The required force shall act against the No. 1 gear. Upon removal of the required force, the maximum permissible gap shall be inspected for compliance with the drawing requirement. Any No. 1 gear and pinion assembly that fails the requirement shall be classed defective and removed from the lot.

4.5.6.2 Load test of NO. 2 sear and pinion assembly. Apply the required axial load perpendicular to the No. 2 gear and in such a direction as to test the assembly of the No. 2 gear and pinion. The required force shall act against the No. 2 gear. Upon removal of the required force, the maximum permissible gap shall be inspected for compliance with the drawing requirement. Any No. 2 gear and pinion assembly that fails the requirement shall be classed defective and removed from the lot.

4.5.6.3 Load test of escapement gear and pinion assembly. Apply the required axial load perpendicular to the escapement gear and in such a direction as to test the assembly of the escapement gear and pinion. The required force shall act against the escapement gear. Upon removal of the required force, the maximum permissible gap shall be inspected for compliance with the drawing requirement. Any escapement gear and pinion assembly that fails the requirement shall be classed defective and removed from the lot.

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4.5.6.4 Load test of lever shaft. Apply the required axial load perpendicular to the lever and in such a direction as to test the assembly of the lever and shaft. The required force shall act against the lever. Upon removal of the required force, the maximum permissible gap shall be inspected for compliance with the drawing requirement. Any lever shaft that fails the requirement shall be classed defective and removed from the lot.

4.5.6.5 Load test of lever pellet. Apply the required axial load perpendicular to the lever and in such a direction as to test the assembly of the lever and the lever pallet. The required force shall act against the lever. Upon removal of the required force, the height of the lever pallet above the lever shall be inspected for compliance with the drawing requirement. Any lever pallet that fails the requirement shall be classed defective and removed from the lot.

4.5.6.6 Torque test of lever shaft. A torque shall be applied between the lever and the lever shaft. Any lever shaft that does not comply with the drawing requirement shall be classed defective and removed from the lot.

4.5.6.7 Load test of annular gear. Apply a load perpendicular to the rotor and in such a direction as to test the assembly of the annular gear studs to the annular gear and the rotor. The required force shall act against the rotor. Upon removal of the required force, the distance from the face of the rotor to the end of stud shall be inspected for compliance with the drawing requirement. Any annular gear that fails the requirement shall be classed defective and removed from the lot.

4.5.6.8 Load test of lock roller pin. Apply the required axial load perpendicular to the rotor and in such a direction as to test the strength of the assembly of the lock roller pin to the rotor. The required force shall act against the rotor. Any lock roller pin that does not comply with the drawing requirement shall be classed defective.

4.5.6.9 Load test of spring stud. Apply the required axial load perpendicular to the rotor housing and in such a direction as to test the assembly of the spring stud and the rotor housing. The required force shall act against the rotor housing. Any spring stud that does not comply with the drawing requirement shall be classed defective.

4.5.6.10 Load test of bridge rivet. Apply the required load perpendicular to the inner plate and into the bridge rivet from the side opposite the bridge. The required force shall act against the inner plate. Any bridge rivet that does not comply with the drawing requirement shall be classed defective.

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4.5.6.11 Load test of pillars. Apply the required axial load perpendicular to the inner plate and in such a direction as to test the assembly of the pillars to the inner plate and outer plate. The required force shall act against the inner plate. Any pillars that do not comply with the drawing requirement shall be classed defective.

4.5.7 Pull-off test of insulator. Secure the setback weight in a holding fixture. An axial pull of a minimum of two pounds shall be applied to the pin assembly. Any insulator that separates from the switch pin shall be classed defective.

4.5.8 pull-off test of switch pin Secure the setback weight in a holding fixture. An axial pull of a minimum of five pounds shall be applied to the pin assembly. Any pin assembly that separates from the setback weight shall be classed defective.

4.5.9 Setback weight return capability. The S&A in the safe position shall be assembled to the test equipment in accordance with 4.4.3. The S&A shall be securely held in place during a 20 pound load test with a hole down force. The 20 pound load shall be applied at an average rate of 300 ± 50 inches per minute to the setback weight. It is mandatory that the setback weight maintain contact with the 20 pound load applicator until fully bottomed. The 20 pound force shall then be removed. S&A'S failing to completely return to the fully safe position shall be classed defective and removed from the lot. Defect standards for verification of the inspection process shall be in accordance with 4.4.4.2 and 4.4.4.3. Defect standards for use with automatic equipment shall be as follows:

a. Units with setback weights which bind when depressed and remain in the depressed condition.

b. Units with setback weights which start to return but do not completely return to the original locking position.

c. Units with setback weights which return from the depressed condition at a slower than normal rate causing rejection.

4.5.10 Rotor reset capability. The S&A shall be assembled to the test fixture in accordance with 4.4.3. The rotor shall be rotated to a position at least 26 degrees, but not more than 35 degrees from the fully safe position and released. Observation shall be made for rotor return to the fully safe position. S&A's failing to comply with the specified requirement shall be classed defective and removed from the lot, This test may be run with the S&A loaded or unloaded.

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4.5.11 Torque test of rotor stop stud (eccentric pin). The required torque shall be applied between the rotor stop stud and gear assembly. The required force shall act on the gear assembly. Any rotor stop stud that does not comply with the drawing requirement shall be classed defective.

4.5.12 Load test of detonator. Apply an axial load of 5 pounds perpendicular to the rotor and in such a direction as to test the strength of the speed clip holding the detonator in the rotor. Any speed clip that moves more than .005 inch shall be classed defective and the S&A removed from the lot.

WARNING: This test may be dangerous and should be performed behind a suitable barricade.

4.5.13 Continuity through detonator test. The S&A shall be armed in the case and the resistance measured between terminals three and four. The value shall be 3.5 ± 1.5 ohms. After the test, the acceptable S&A's shall be disarmed and returned to the line for further assembly Any S&A failing the requirement shall be classed defective and removed from the lot.

CAUTION: Any continuity measurement which passes electrical current through the detonator shall be limited to one milliampere maximum.

4.5.14 Temperature conditioning. Prior to the centrifuge test (4.5.15 and 4.5.16), the sample shall be conditioned for a minimum of two hours after temperature stabilization at the following temperatures:

70° ± 15°F - 15 units

-50° ± 5°F - 15 units

+150° ± 5°F - 15 units

4.5.15 M439 S&A non-arming test. The S&A in the safe position and in the horizontal plane with the G-weight located in the downward direction (closest to earth), shall be assembled to the test equipment in accordance with 4.4.3. The equipment shall accelerate the loaded Safety and Arming Device in a clockwise direction to 12.0 g's minimum for a minimum of five (5) seconds. Observation shall be made for rotor in the safe position. S&A' S failing to comply with the specified requirement shall be classed defective and removed from the lot.

4.5.16 M439 S&A arming time test. The S&A in the safe position and in the horizontal plane with the G-weight located in the downward direction (closest to earth), shall be assembled to the test equipment in accordance with 4.4.3. The equipment shall

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accelerate the S&A to 55 g's ± 0.5 g's in a direction which simulates rocket launch. The arming time shall be recorded to the nearest hundredth of a second using a calibrated timer. At the completion of acceleration, the equipment shall be halted and visual observation shall be made for the locking position of the detent; to ascertain positive arming. At the completion of the test, the S&A shall be returned to the safe position. Any S&A that fails the requirement shall be classed defective and removed from the lot.

WARNING : This test may be dangerous, and should be performed behind a suitable barricade.

4.5.17 Module time delay function test. (See Figure 1) A compensated voltage setter that represents the helicopter setter system shall be used. After shorting pins 5, 6 and 4 together for seven seconds minimum, the module shall be capable of accepting a setting charge from the compensated voltage setter and exhibiting a corresponding time delay as listed in Table VI. The time delay shall be the time interval from the time pin 2 is connected to pin 4 and the leading edge of the output pulse exits pin 3. The firing output pulse (pin 3) shall provide at least 1000 ergs of energy in less than 100 microseconds into an M84 detonator or a resistor of 3.5 ± 1.5 ohms. Record the time delay setting, function time delay and energy output (equal to, greater than or less than by serial number or module lot no. Any module that does not meet the requirement shall be classed defective and removed from the lot.

NOTE: Inadequate discharge of residual voltages could affect the timing accuracy of the module. In certain cases, it may require a second discharge operation. Therefore, the contractor shall develop procedures to ensure that the residual voltage of the timing and firing capacitors is equal to or less than 0.005 volt when checked 14 days after final discharging.

4.5.17.1 Module time delay function test. (After tear down from centrifuge lot Acceptance.) The module shall function within the requirements of Table III (+70°F only). If two or more do not meet the requirements, the lot shall be rejected.

4.5.18 Module thermal shock. The module shall be subjected to the Thermal Shock Test in accordance with MIL-STD-331, Test C7. These modules shall then be subjected to the timing test as specified in 4.5.18. Any module that does not meet the test requirements shall be classed defective. Two defects shall be cause for lot rejection.

4.5.19 S&A position check. The S&A position, before insertion into the fuze, shall be inspected. Any armed S&A shall be considered defective and removed from the lot, and the

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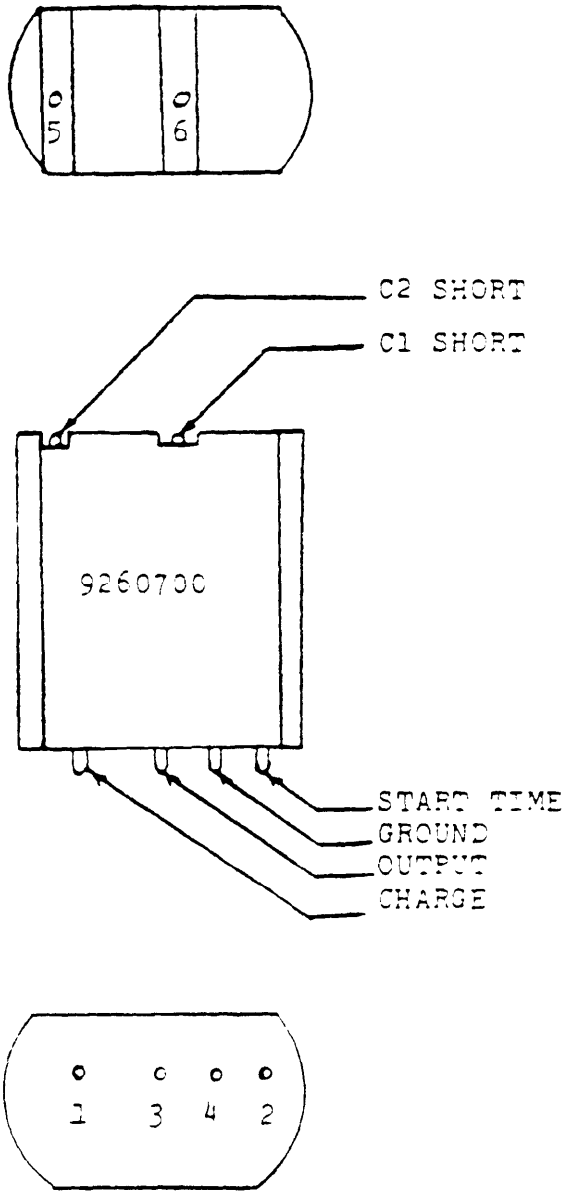


FIGURE 1 Electronic module terminal diagram.

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TABLE III. Module time delay test.

1. All modules shall be serialized.

2. A setter that represents the helicopter setter system shall be used for acceptance of all modules.

Nominal Time Settings	Environment	Module Quantity	Acceptance Limits (sec)
2 Sec.	-50°F	50	1.87 to 2.07
	+70°F	50	1.87 to 2.07
	+150°F	50	1.87 to 2.07
	Thermal Shock	12	1.87 to 2.07
10 Sec.	-50°F	50	9.80 to 10.25
	+70°F	50	9.80 to 10.20
	+150°F	50	9.80 to 10.25
	Thermal Shock	12	9.80 to 10.20
24 Sec.	-50°F	50	23.52 to 24.60
	+70°F	50	23.52 to 24.48
	+150°F	50	23.52 to 24.60
	Thermal Shock	12	23.52 to 24.48

The firing capacitor shall be checked to insure that at least minimum detonator firing output pulse is present when the detonator firing circuit is tripped at the conclusion of the time delay.

3. The module timing circuit shall be tripped immediately following the charging of each module.

4. Modules shall be temperature conditioned for a minimum of two (2) hours after stabilization at the temperatures specified in paragraph 2 above.

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assembly operation shall be halted until operation is corrected to Government satisfaction. The inspection shall be performed using procedures in accordance with 3.16, 4.4.4.2 and 4.4.4.3 and equipment in accordance with 4.4.3. Defect standards with semi-armed and fully armed conditions shall be introduced at various times to verify test performance. The Design Agency approved handling procedure to preclude untested S&A's from being introduced into the fuze lot shall also be assessed when the standards are introduced. If an improper procedure is being used as determined by the Government, all fuzes accepted subsequent to the last acceptable handling procedure assessment shall be screened as determined by the government.

4.5.20 Charge circuit continuity. The fuze shall be subjected to a charge circuit continuity test. Any fuze which is identified as a defect shall be removed from the lot or sample and held for failure analysis or rework. An open circuit continuity failure may indicate that the safety and arming device is armed; any fuze with this type of defect should be handled with appropriate safety precautions. The test shall be conducted using Government approved equipment and procedures.

NOTE: No more than three (3) continuity tests shall be performed within a 24-hour period prior to any time delay function test. Excessive continuity tests could result in a buildup of residual charge in module. A buildup of residual charge could affect the accuracy of the timing test.

4.5.21 Temperature and humidity. The fuze shall be subjected to the Temperature and Humidity Test per MIL-STD-331, Test C1 with the exception that only one 14 day cycle be used. Any fuze that does not meet MIL-STD-331 test requirements shall be classed defective.

4.5.22 Thermal shock. The fuze shall be subjected to the Thermal Shock Test in accordance with MIL-STD-331, Test C7. These fuzes shall then be centrifuge fired. The fuze shall be charged to 20.275 seconds prior to arming. After arming, the fuze shall be fired. Any fuze that does not meet test requirements of 4 .5.35.2 shall be classed defective.

4.5.23 Jolt and jumble. The fuze shall be subjected to the Jolt Test specified in MIL-STD-331 with the exception that the fuze shall not be disassembled and inspected after the Jolt Test. After completion of the Jolt Test, the same fuze shall be subjected to and withstand the Jumble Test specified in MIL-STD-331. Any fuze that fails the requirements shall be cause for lot rejection.

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4.5.24 12 Meter (40-foot) drop. The fuze with test warhead and inert motor P/N 233AS431 shall be subjected to the 12 Meter Drop Test as specified in MIL-STD-331. The complete pack as specified on dwg. 9230116 and 9235840 shall be used. Any fuze that fails the requirement shall be cause for lot rejection.

4.5.25 One and one-half Meter (5-foot) drop. A random sample of fifteen (15) fuzes shall be selected and subjected to the test as specified in MIL-STD-331 with the following conditions:

a. The fuze will be assembled to a test Warhead. Three (3) fuzes must be dropped in each of the five orientations. No fuze shall be subjected to more than one drop. The orientations of the Safety and Arming Device (specifically the setback weight) of the fuze shall be random.

b. The One and one-half Meter Drop Test, as specified in MIL-STD-331, shall be performed until three (3) consecutive lots (a minimum lot size of one week's production) have complied with the acceptance criteria specified. After satisfying the above requirement, one lot (a minimum lot size of one week's production) per month shall be selected by the Government and subjected to and pass the Five Foot Drop Test. Should the one lot (per month) fail the acceptance criteria, the contractor shall be required to return to the three (3) consecutive lot procedures above for process qualification.

c. The fuzes shall be subjected to the tests and acceptance or rejection criteria as specified in Table IV. Fuzes must be safe to handle and safe to fire or dispose of (not necessarily operable) following the One and one-half Meter Drop Test and prior to performing the Non-arm and Arming Time Tests.

TABLE IV. Acceptance criteria, five foot drop test.

Test	Accept	Reject	Test Method Paragraph
Non-Arm	0	1	4.5.29
Arming Time Under min.	0	1	4.5.30

Any fuze that is unsafe to handle (See 6.7.1), fire or dispose of following the Five Foot Drop Test or does not meet the arming time under minimum definitions of 6.7.2 is classified as a safety failure. If the sample fails the above acceptance criteria or has one or more safety failures, the lot shall be rejected.

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4.5.26 Non-environmental test. A random sample of twelve (12) fuzes shall be selected and subjected to the test and acceptance or rejection criteria as specified in 4.5.28.

4.5.27 Helicopter vibration test. Fuzes shall be subjected to Helicopter Vibration as specified in Figure 2. Vibration conditioning at resonance is not required. Following vibration conditioning, the fuzes shall be subjected to the tests and acceptance or rejection criteria specified in 4.5.28.

NOTE: The fuzes shall be subjected to and must pass the continuity test (4.5.20) subsequent to the vibration test. If one or more continuity failures occur in the test sample the lot shall be rejected.

4.5.28 Acceptance criteria. The sample fuzes from the Non-environmental (12) and Environmental tests (20) shall be combined into one sample and subjected to the tests specified in Table V and either Figure 3 or Figure 4, whichever is applicable. Arming time definitions are in 6.7.2.

TABLE V. Acceptance criteria, non-environmental and vibration tests.

Test	Accept	Reject	Test Method Paragraph
Non-arm	0	1	4.5.29
Arming Time under min.	0	1	4.5.30
Arming Time over max.	0	1	4.5.30
Fail to arm	0	1	4.5.30
Mechanical Interrupt	0	1	4.5.31
Charge Circuit Continuity	0	1	4.5.20

4.5.29 M439 fuze non-arming test. The fuze in the horizontal plane with the G–weight located in the downward direction (closest to earth), shall be assembled to the test equipment in accordance with 4.4.3. The equipment shall accelerate the loaded fuze in a clockwise direction to 12.00 g’s plus or minus 0.5 g’s for a minimum of three (3) seconds. An examination shall be made for rotor in the safe position. Any fuze that fails the requirements of the non-arming test shall be classed defective.

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One 15 minute sinusoidal sweep (5-500-5 HZ) shall be performed for each of three axes. No dwell time shall be spent on any one frequency. A search for resonance shall not be done.

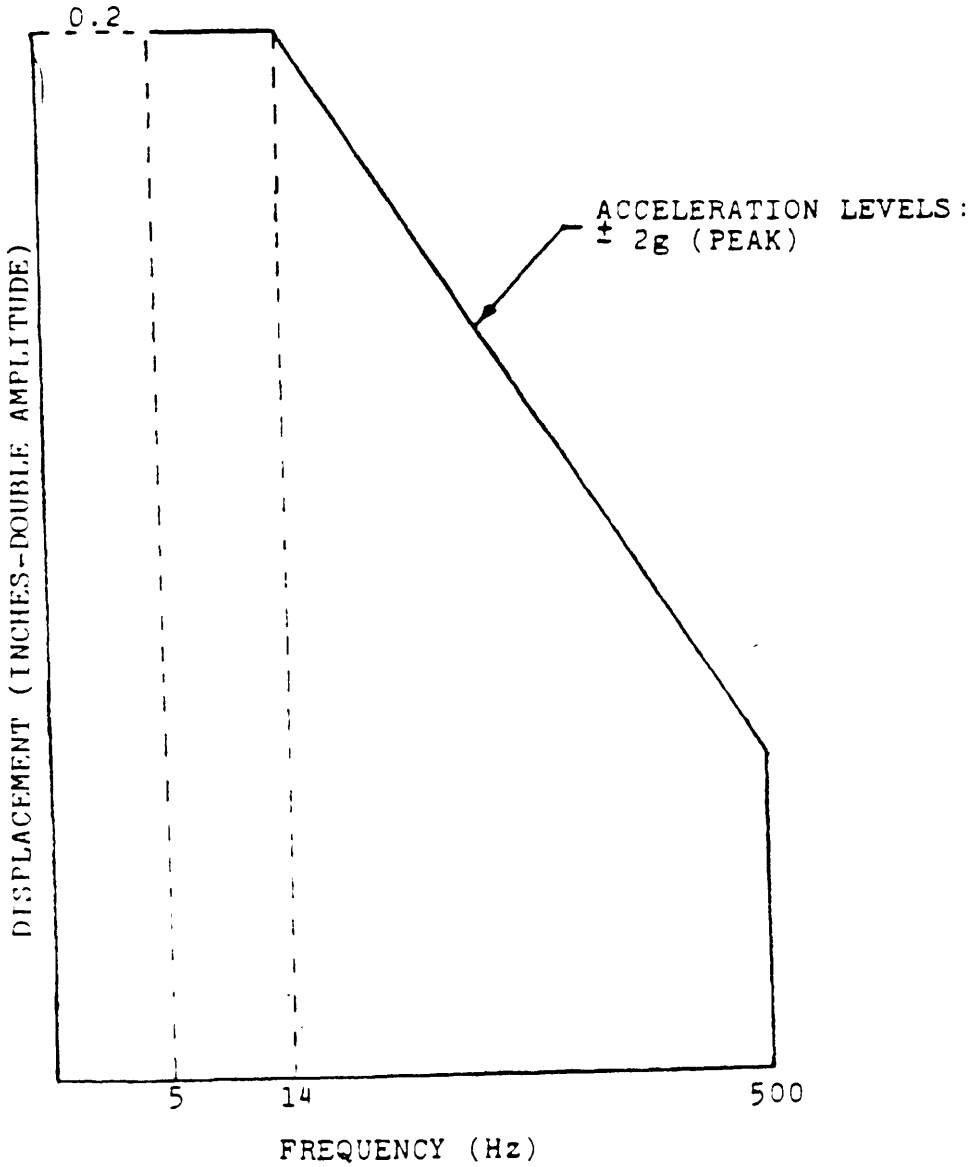
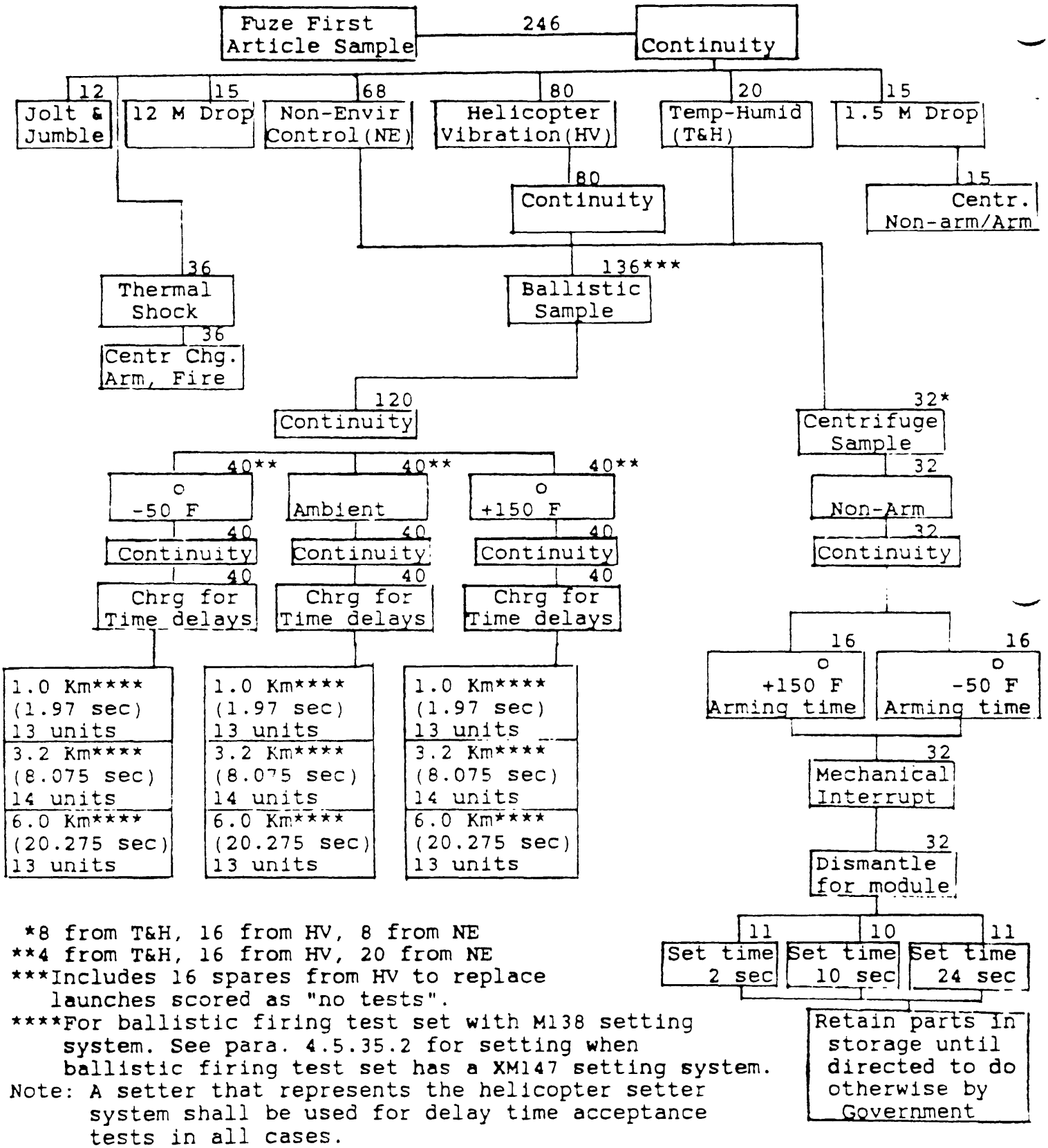


FIGURE 2 Helicopter vibration.

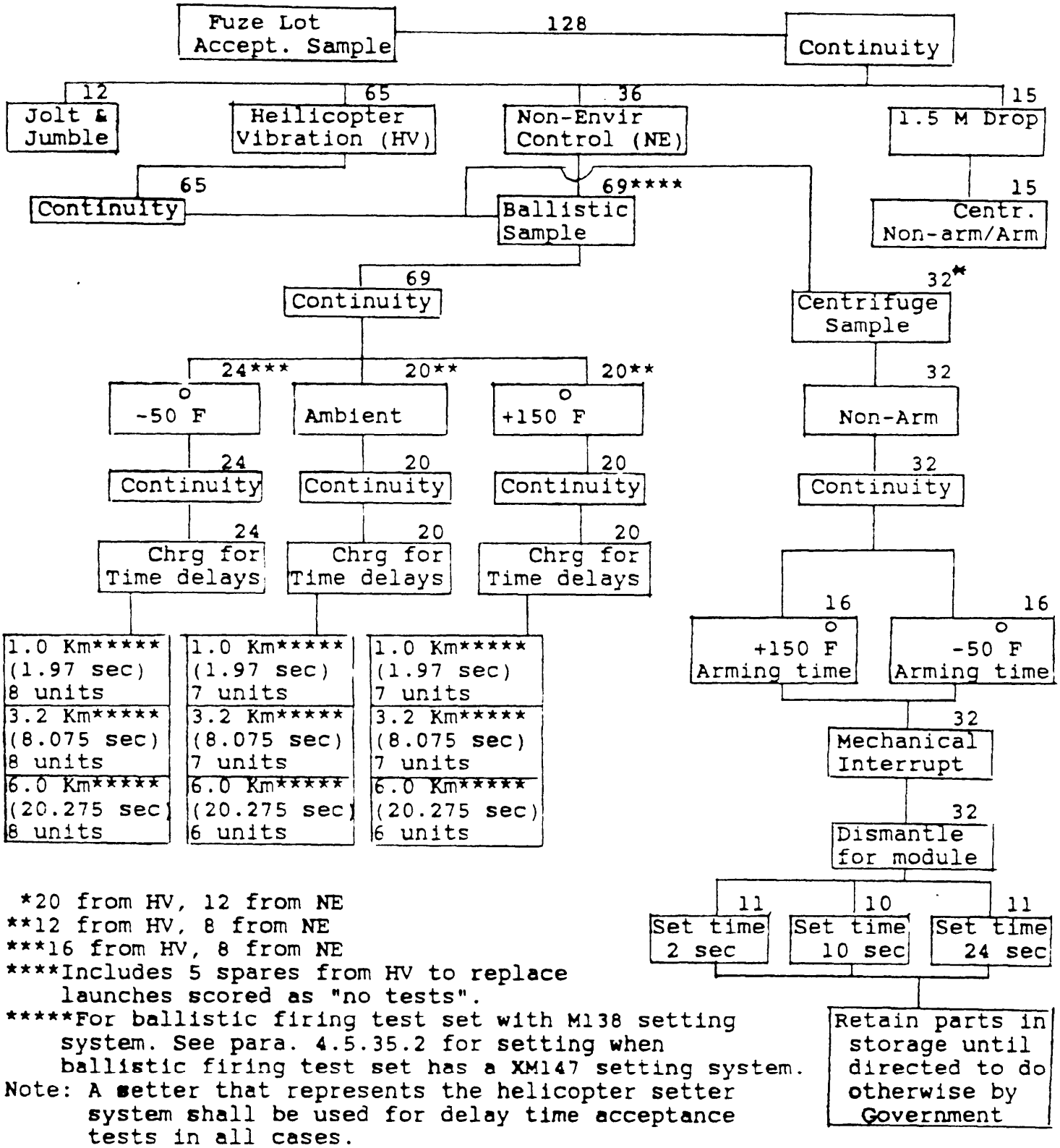
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*8 from T&H, 16 from HV, 8 from NE
**4 from T&H, 16 from HV, 20 from NE
***Includes 16 spares from HV to replace launches scored as "no tests".
****For ballistic firing test set with M138 setting system. See para. 4.5.35.2 for setting when ballistic firing test set has a XM147 setting system.
Note: A setter that represents the helicopter setter system shall be used for delay time acceptance tests in all cases.

FIGURE 3. First article test.

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*20 from HV, 12 from NE
**12 from HV, 8 from NE
***16 from HV, 8 from NE
****Includes 5 spares from HV to replace launches scored as "no tests".
*****For ballistic firing test set with M138 setting system. See para. 4.5.35.2 for setting when ballistic firing test set has a XM147 setting system.
Note: A setter that represents the helicopter setter system shall be used for delay time acceptance tests in all cases.

FIGURE 4 Fuze lot acceptance test.

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4.5.30 M439 fuze arming time test. The fuze in the horizontal plane with the G-weight located in the downward direction (closest to earth), shall be assembled to the test equipment in accordance with 4.4.3. The equipment shall accelerate the loaded fuze in a clockwise direction to 55.0 ± 0.5 g's and the arming time shall be recorded to the nearest hundredth of a second using a swing fixture and acoustic pick-up feeding the arming cycle noise signal to a design agency approved recorder. The arming time as recorded using the swing fixture shall be adjusted using the correlation factor as determined in 4.5.3.4. Any fuze that fails the requirements of the arming time test shall be classed defective.

4.5.31 Mechanical interrupt test. The armed fuzes from 4.5.30 shall be checked for charge circuit continuity using Government approved equipment and procedures. Any fuze that does not indicate an open circuit between the charge circuit and the umbilical shall be classed defective.

4.5.32 Test equipment failure. If the test equipment fails to produce an arming readout (i.e., recording paper skewes, audio pick-up, or amplifier inoperative) during the arming time test as verified by the Government, the Safety and Arming Device shall be visually inspected for arming; if found to be armed or partially armed, score as a "no test" and replace with a spare fuze. The timing record produced by the fuze on the approved fuze timing equipment shall be considered the timing record of the fuze under the applicable defect classification.

If the electronic module time delay test equipment is suspected to have malfunctioned during the module time delay testing specified of 3.14. The assembly may be tested four (4) more times and if all time delays meet the acceptance limits, the first reading may be considered as a no test and the second reading shall be used Procuring contract officer's approval must be obtained prior to the retest.

4.5.33 Rounding off of fuze function times, fuze arming times and module function times. Fuze function times, fuze arming times, and module function times if recorded to more than two (2) places to the right of the decimal point shall be "rounded" off to the hundredth of a second.

4.5.34 Correlation factor. A correlation factor, which correlates the fuze arming time method (swing fixture) to the in-line S&A arming time method, shall be run for this fuze. Correlation tests shall be performed by the contractor with Government verification. Correlation factors shall be validated at six month intervals (maximum) and re-established whenever:

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a. A new contract is issued

b. A change of fuze type

c. Fuze timing equipment is modified

For the correlation procedure a digital counter or an approved equivalent device shall be used, set to read milliseconds for S&A timing and set for oscillograph traces, to the nearest millisecond, for the fuze. The G-weight shall be located in the downward direction (closest to earth) for both tests. The correlation shall be accomplished as follows:

a. Select five (5) S&A's demonstrating acceptable arming time . Each S&A shall have repeatable arming times within ± 25 milliseconds in five successive runs on the approved S&A timing equipment. S&A arming time readings shall be obtained from the first movement of the setback weight to the detent locking in the rotor.

b. The same S&A's shall be assembled to electronic modules and placed in fuzes. The fuzes shall then be timed five (5) times on the approved fuze timing equipment. Readings of the fuze arming time traces, from setback weight bottoming to rotor detent, shall be recorded.

c. The shift in average fuze time, if any, when compared with the average S&A time (taken as a standard) shall be used as the correlation correction and applied to the acceptance test results.

4.5.35 Ballistic firinig test. The loaded M439 fuze shall be conditioned in accordance with Figure 3 or Figure 4, whichever is applicable. This test shall be performed at a Government facility using PXR6358 test warheads. MK66 rocket motors shall be used for all tests. The fuze shall be set with a Government approved Ballistic Firing Test Set. A hard copy of the fuze charge signature shall be made at this time. Prior to setting arid firing, the fuzes shall be checked for charge circuit continuity (4.5.20). Fuzes that fail the continuity test shall not be subjected to the ballistic test. The fuze shall be removed from the test vehicle and checked for continuity to determine fuze or test vehicle failure. If it is determined that there is a test vehicle failure then it is considered a no-test. Firing and fuze charging shall be accomplished by the ballistic firing test set using fuze time settings of 2, 8, and 20 seconds. Time from launch to fuze function, including the possibility of function upon S&A arming, shall be detected and recorded.

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4.5.35.1 No-test conditions. Some ballistic firings may result in conditions that do not properly test fuze time delay function. If the following conditions occur during a test firing, that launch will be scored as a "no test."

a. The rocket hits the ground before the maximum time delay for that time setting occurs.

b. The fuze is not charged properly due to malfunctioning of the setter or the connecting cables.

4.5 .35.2 Acceptance criteria. There are two different ballistic firing test set configurations used for the ballistic firing test. one contains a M138 helicopter setting system and car, be recognized by the red illuminated display panel when the unit is powered up. The other contains a XM147 helicopter setting system and can be recognized by the blue-green illuminated panel when the unit is powered up. The Table VI below lists the range settings (nominal set times) and acceptance criteria for each of the units.

TABLE VI. Function time criteria.

Nominal Time Setting (Sec.)	Function Time Delay (Sec.)
M138 Red panel illumination	
1.0 Km (1.975)	1.85 to 2.09
3.2 Km (8.075)	7.87 to 8.27
6.0 Km (20.275)	19.77 to 20.78
XM147 Blue Green panel illumination	
1.2 Km (2.05)	1.93 to 2.17
3.7 Km (8.01)	7.81 to 8.21
6.6 Km (20.50)	20.00 to 21.00

4 .5.35.3 Rejection criteria.

4.5.35.3.1 First article sample. If a total of four (4) or more units do not meet the function time delays of 4.5.35.2 or fail the charge circuit continuity test, the sample shall be rejected. Also, if one (1) or more failures are functions upon S&A arming, or if three (3) or more failures are duds, the sample shall be rejected. Functions upon S&A arming are failed units that functioned at less than 1 sec after motor ignition. Duds are failed units other than those that do not meet the function time delays or fail the charge circuit continuity test.

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4 .5.35.3.2 Lot acceptance sample. If a total of five (5) or more units do not meet the function time delays of 4.5.35.2 or fail the charge circuit continuity test, the lot shall be rejected. Also, if one (1) or more failures are functions upon S&A arming, or if three (3) or more failures are duds, the lot shall be rejected. Functions upon S&A arming are failed units that functioned at less than 1 sec after motor ignition. Duds are failed units other than those that do not meet the function time delays, or fail the charge circuit continuity test.

4.5.36 Data items. The contractor shall submit a test/inspection report in accordance with 6.3 giving the detailed test results.

5. PACKAGING

5.1 Packaging. The fuzes shall be packaged as specified on Dwg . 9357775. Or , as alternate, packaging, packing and marking shall comply with CFR49, Parts 100-199.

5.2 Packing. Fuzes shall be packed in accordance with Dwg. 9357776.

5.3 Marking. Marking shall be in accordance with Dwg. 8799112.

5.4 Ammunition data cards. Data card information shall be as specified in MIL-STD-1167.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The fuze covered by this specification is intended for use as part of the 2.75 Inch Rocket System.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).

c. Requirements for submission of first article sample.

d. Applicable stock number.

e. Packaging requirements. if other than specified in Section 5.

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f. Serialization requirements, if applicable.

g. Certificate of conformance for each lot or shipment of product.

6.3 Data requirements. The following data requirement shall be developed when this specification is applied on a contract, per the Contract Data Requirements List, DD Form 1423.

Reference Para.	DID Number	DID Title
4.5.36	DI-NDTI-80809A	Test/Inspection Report

6.3.1 Defect reporting. From each lot produced, the Contractor shall submit to Commander, USA ARDEC, ATTN: SMCAR-QAF-A, Picatinny Arsenal, NJ 07806-5000, results of the Module, Jolt, Jumble, Helicopter Vibration, and Centrifuge tests within five (5) working days after completion of acceptance testing of the lot. If tests in addition to acceptance requirements are utliized to evaluate the lot, results of these tests will also be submitted within five (5) working days after the additional tests are completed. The defective fuzes shall be saved pending review of the results. See Figure 5 for example of test report format.

6.3.2 Correlation factor. Data obtained in accordance with 4.5.34 shall be submitted to Commander USA ARDEC, ATTN: SMCAR-QAF-A, Picatinny Arsenal, NJ 07806-5000.

6.4 Submission of inspection equipment for design approvals. Submit copies of designs as required to: Commander, U.S. Army ARDEC, ATTN: SMCAR-QAF-A, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.5 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament, Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as U.S. Army Armament, Research, Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal, Missile Command (MICOM), or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.6 Submission of alternative quality conformance provisions. All contractor proposed alternative quality conformance provisions will be submitted to the Government for evaluation/approval as directed by the contracting activity.

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M439 METAL PARTS LOT ACCEPTANCE TEST DATA:

Contractor: _____ Fuze Lot No.: _____

Date of Test

Start: _____ Lot Quantity (Gross): _____

End: _____ Sample Size (Total) _____

Sample Size _____

Returned to Lot: _____

Date Shipped: _____ Lot Quantity (Net): _____

One and one-half Meter (5-foot) Drop

sarnple Size	Serial No. of Defect	Failure Analysis

Non-Arm

Sample Size	Serial No. of Defect	Failure Analysis

FIGURE 5. Test report.

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Arming Time

Sample	Number Accepted	Serial No. of Short Times	Serial No. of Long Times	Timed but Failed to Arm	Not Timed and failed to Arm	Armed but not Timed

Attach list of fuze serial numbers and arming times.

Operational Tests

Vibration Test

Sample Size	Serial No. of Defects	Failure Analysis

FIGURE 5. Test report. (continued)

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Fuze Non-Arm and Arming-Time Tests

Non-Arm Test

Sample Size	Serial No. of Defects	Failure Analysis

Arming Time Test

Sample Size	Number Accepted	Serial No. of Short Times	Serial No. of Long Times	Timed but Failed to Arm	Not Timed and failed to Arm	Armed but not Timed

Attach list of fuze serial numbers and arming times.

Failure Analysis

Correlation applied to the arming time readings of the One and one-half Meter (5-foot) Drop and the Operational Test samples was _____ seconds.

FIGURE 5. Test report. (continued)

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Module Time Delay Tests

Sample Size	Number Accepted	Serial No. of Short Times	Serial No. of Long Times	Serial No. of Failures to Function	Serial No. of Low Outputs

Jolt and Jumble

Sample Size	Serial No. of Defects	Failure Analysis

Failure Analysis

If lot failure is due to other than test conditioning.

Government Inspector Date

Contractor Representative Date

FIGURE 5. Test report. (Continued)

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6.7 Definitions.

6.7.1 Safe to handle. The fuze will be considered safe to handle if after electronic examination and tear down the following conditions are met:

a. The rotor shall be in the safe position and secured by setback weight such that it cannot be turned to the armed position with finger pressure.

b. The lock roller pin on the rotor shall not be broken.

c. The setback weight shall be in the fully safe position (raised to its highest position).

d. No loose explosive shall be present in the fuze assembly.

e. No parts shall be broken or deformed which. could present a safety hazard in handling or inadvertent firing.

6.7.2 Fuze arming and non-arming definitions.

a. Times under 0.70 second are classed as arming time under minimum.

b. Times 0.91 second through 1.20 seconds are classed as arming time over maximum.

c. Times over 1.20 seconds or fuzes which do not arm or fuzes which time but do not detent are classed as failures to arm.

6.8 Contractor furnished parts.

6.8.1 Packaging and packing. The contractor will provide the paackaging and the packing for the 12 Meter (40-foot) Drop.

6.8.2 PXR6358 test warhead. The contractor will provide PXR6358 test warheads for the 12 Meter (40-foot) Drop, One and one-half Meter (5-foot) Drop and Ballistic Tests.

6.8.3 Inert MK66 rocket motor. The contractor will provide Inert MK66 Rocket Motor for the 12-meter (40-foot) Drop Test.

6.9 Source control and specification control drawings. Certificates of Conformance from the supplier certifying that the item compiles with the drawing requirements shall be provided to the Government representative upon request.

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6.10 Subject term (key word) listing.

Hydra 70
2.75 Inch
Multi-purpose Submunition

6.11 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army - AR

(Project 1340-A877)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and re letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification o requirements on current contracts. Comments submitted on this form do not constitute or imply authorization any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1 DOCUMENT NUMBER	2 DOCUMENT DATE (YYMMDD)
	MIL-F-48877 (AR)	941013

3 DOCUMENT TITLE FUZE, ROCKET: M439 PARTS AND ASSEMBLY WITH DETONATOR

4 NATURE OF CHANGE (Identify paragraph number and include proposed rewrite if possible. Attach extra sheets if needed.)

5 REASON FOR RECOMMENDATION

6 SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8 PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
U.S. ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial 201-724-6675	(2) AUTOVON DSN-880-6675
c. ADDRESS (Include Zip Code) ATTN: AMSTA-AR-EDE-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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